

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S(TEPS)]

PAPER-1

Time : 3 Hours

Maximum Marks : 222

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains 28 pages and that all the 18 questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. Do not use the ORS for rough work.
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. Do not write any of these details anywhere else on the ORS. Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a BLACK BALL POINT PEN to darken the bubbles on the ORS.
11. Darken the bubble  COMPLETELY.
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.
15. Take $g = 10 \text{ m/s}^2$ unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- Boltzmann constant $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
- Coulomb's law constant $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- Universal gravitational constant $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- Speed of light in vacuum $c = 3 \times 10^8 \text{ ms}^{-1}$
- Stefan-Boltzmann constant $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- Wien's displacement law constant $b = 2.89 \times 10^{-3} \text{ m-K}$
- Permeability of vacuum $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- Permittivity of vacuum $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- Planck constant $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

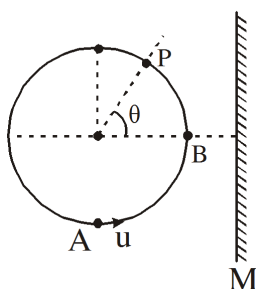
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

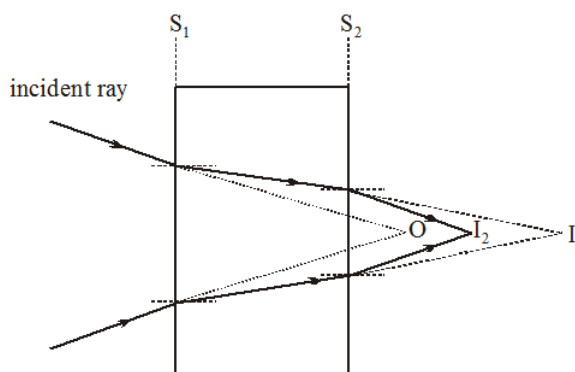
1. A particle is moving along a circle with a constant speed u , as shown in figure. M is a plane mirror, then



- (A) relative velocity of image w.r.t. object becomes zero twice in one revolution
 (B) path of image with respect to object is circle
 (C) path of image with respect to object is straight line
 (D) speed of image with respect to object when particle is at P is $2u \sin \theta$

Space for Rough Work

2. The image of an object kept at a distance 20 cm in front of a concave mirror is found to coincide with itself. If a glass slab ($\mu = 1.5$) of thickness 3 cm is introduced between the mirror and the object, then in order that the final image again coincides with the object :
- (A) the mirror should be displaced away from the object
 (B) the mirror should be displaced towards the object
 (C) the magnitude of displacement is 1 cm
 (D) the magnitude of displacement is 0.5 cm
3. Figure shows converging rays on a glass slab placed in air. O is point of intersection of incident rays. I_1 and I_2 are image formed after refraction at S_1 and S_2 surface respectively. Mark the correct option(s)



- (A) for refraction at S_2 , I_1 is real object and I_2 is real image
 (B) for refraction at S_1 , O is real object and I_1 is its real image.
 (C) for refraction at surface S_1 , O is virtual object and I_1 is real image
 (D) for refraction at surface S_2 , I_1 is virtual object and I_2 is real image
4. Consider a shell that has a muzzle velocity of 45 ms^{-1} fired from the tail gun of an airplane moving horizontally with a velocity of 215 ms^{-1} . The tail gun can be directed at any angle with the vertical in the plane of motion of the airplane. The shell is fired when the plane is above point A on ground, and the plane is above point B on ground when the shell hits the ground. (Assume for simplicity that the Earth is flat)
- (A) Shell may hit the ground at point A.
 (B) Shell may hit the ground at point B.
 (C) Shell may hit a point on earth which is behind point A.
 (D) Shell may hit a point on earth which is ahead of point B.

Space for Rough Work

5. A string of length 1m and linear mass density 0.01 kg/m is stretched to a tension of 100 N. When both ends of the string are fixed, the three lowest frequencies for standing wave are F_1 , F_2 and F_3 . When only one end of the string is fixed, the three lowest frequencies for standing wave are n_1 , n_2 and n_3 . Then

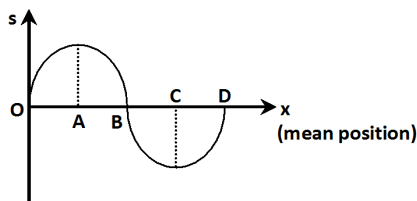
(A) $n_3 = 5 n_1 = F_3 = 125 \text{ Hz}$

(B) $F_3 = 5 F_1 = n_2 = 125 \text{ Hz}$

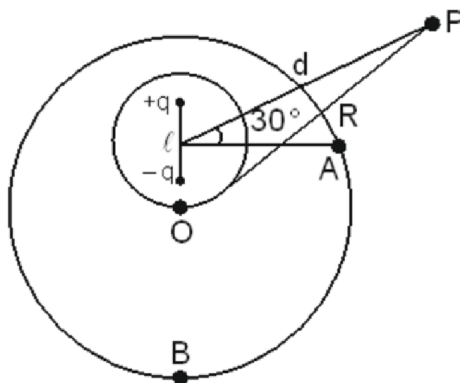
(C) $F_3 = 2n_2 = 3F_1 = 150 \text{ Hz}$

(D) $n_2 = \frac{F_1 + F_2}{2} = 75 \text{ Hz}$

6. A sound wave is travelling in air along negative x-direction. Displacement (s) of particles from their mean positions at a particular time t is shown in the figure. Choose the correct option(s) for the instant only.



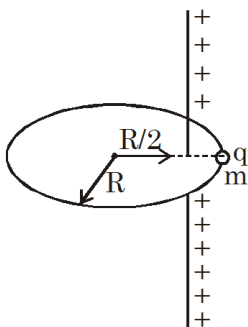
- (A) Particle located at C has zero velocity
 (B) Particle located at D has its velocity in the negative direction
 (C) The pressure at C is equal to normal atmospheric pressure
 (D) Particles located near B are under compression
7. A cavity is taken out from a uniform conducting sphere. Inside the cavity a dipole is placed as shown in the figure. Find which of the following is/are true ? (Given : $q = 10^{-1} \text{ C}$, $\ell = 0.1 \text{ mm}$, $\theta = 30^\circ$, $d = 10 \text{ cm}$, $R = 12 \text{ cm}$)



- (A) Potential at points A and B are same
 (B) Potential at points A and B are different
 (C) Potential at point P is 0 volt.
 (D) Potential at point P is 2 volt.

Space for Rough Work

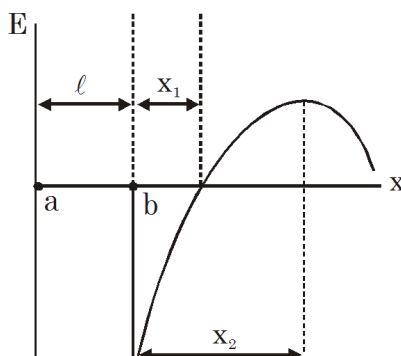
8. A bead of charge $+q$ and mass m is threaded on a smooth fixed horizontal circular frame of radius R . An infinite vertical line charge of charge/length λ passes through its plane at a distance $\frac{R}{2}$ from the center. The bead is slightly displaced and it moves on the circular path. Choose the correct statement(s) :-



- (A) The maximum speed of the bead is $\sqrt{\frac{q\lambda \ell n 3}{2\pi m \epsilon_0}}$
- (B) The maximum speed of the bead is $\sqrt{\frac{q\lambda \ell n 3}{\pi m \epsilon_0}}$
- (C) The initial normal force on bead is $\sqrt{\left(\frac{q\lambda}{\pi \epsilon_0 R}\right)^2 + m^2 g^2}$
- (D) The normal force in radial direction at maximum speed is $\frac{q\lambda}{\pi \epsilon_0 R} \left[\ell n 3 + \frac{1}{3} \right]$

Space for Rough Work

9. Two point charges are placed at point a and b. The field strength to the right of the charge Q_b on the line that passes through the two charges varies according to a law that is represented graphically in the figure. Find the signs of the charges and ratio of magnitudes of charges Q_a/Q_b and the distance x_2 , of the point from b where the field is maximum, in terms of ℓ and x_1 .



- (A) Here Q_a is positive and Q_b is negative
- (B) Here Q_a is negative and Q_b is positive
- (C) $\left| \frac{Q_A}{Q_B} \right| = \left(\frac{\ell + x_1}{x_1} \right)^2$
- (D) $x_2 = \frac{\ell}{\left(\frac{\ell + x_1}{x_1} \right)^{2/3} - 1}$
10. A small light source moves at a constant speed v along a straight line making an angle 10° with the main optical axis of the converging lens. Image of the source also moves along straight line, which is at an angle 20° with the main optical axis of the lens. Find the minimum image speed (u_{rel}) relative to the source. What's the magnification at this moment ?
- (A) $u_{\text{rel}} = \frac{v}{2}$
- (B) $u_{\text{rel}} = v \frac{\sqrt{3}}{2}$
- (C) $m = -0.5$
- (D) Image must be real

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

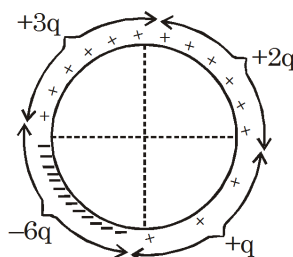
- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct answer is chosen.

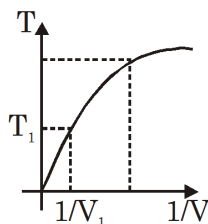
Zero Marks : 0 In all other cases.

1. A charged non conducting ring has quadrants with different charges uniformly distributed as shown.

If $\alpha \times \frac{qR\sqrt{34}}{\pi}$ is net dipole moment of this system then find the value of α .

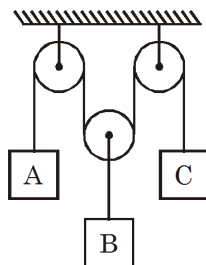


2. A racing car start from rest with acceleration 'a' on a straight level track towards a boy standing on the track at distance 2000 m. At an instant 22 sec after the car starts the observed frequency is 1.5 kHz while actual frequency of horn is 1 kHz. If the speed of sound is 330 m/s then write 'a' (in m/s^2). Neglect the time taken by sound to reach the body.
3. Figure shows a parabolic graph between T and $\frac{1}{V}$ (T = Temperature, V = Volume) for a gas undergoing an adiabatic process. If the ratio of rms velocity of molecules and speed of sound in the gas at the same temperature is $\sqrt{n}$, then find the value of n.



Space for Rough Work

4. In shown system strings and pulley are frictionless and massless. Block A is moving downward with velocity of 7 m/s , which is increasing at a rate of 2 m/s^2 , while block C is moving upward with a velocity of 9 m/s , which is decreasing at a rate of 3 m/s^2 . The block B is moving downward with velocity of $v_B\text{ m/s}$, which is decreasing at a rate of $a_B\text{ m/s}^2$. Find the value of $\frac{2a_B}{v_B}$.



5. A concave spherical mirror produces an inverted image of an object. The distance between object and image is $d = 12\text{ cm}$ and lateral magnification is -2 . Find focal length of mirror in cm.
6. A vessel, whose bottom is flat and perfectly reflecting, is filled with water (index $= 4/3$) upto a height $= 40\text{ cm}$. A point object in air above is moving towards the water surface with a constant speed $= 4\text{ m/s}$. What is the relative speed of its final image (in m/s), as seen by the object itself, at a moment when the object is 30 cm above the water surface ?

Space for Rough Work

SECTION-IV : (Maximum Marks: 16)

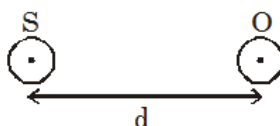
- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen

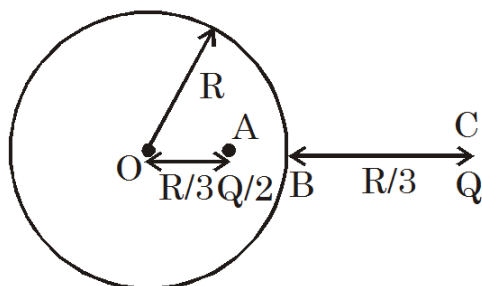
Zero Marks : 0 In all other cases

1. In column-I, certain situation are indicated. In Column-II, comments about intensity/frequency detected by observer are indicated. Here f_0 & I_0 are intensity recorded in static situation (when both source & observer are at rest). In all cases, d is distance when observer detects frequency & intensity.



	Column-I		Column-II
(A)	Source emits plane wavefront and moving towards stationary observer	(P)	$f > f_0, I > I_0$
(B)	source emits cylindrical wavefront and moving towards stationary observer	(Q)	$f > f_0, I = I_0$
(C)	source emits spherical wavefront and moving towards stationary observer	(R)	$f > f_0, I < I_0$
(D)	Observer moving towards stationary source emitting spherical wavefront	(S)	$f < f_0, I = I_0$
		(T)	$f < f_0, I < I_0$

2. We have a thin neutral conducting shell of radius R inside which we kept a charge $\frac{Q}{2}$ & outside a charge Q . The centre of shell, $\frac{Q}{2}$ & Q are all on same straight line. Match options in column-II with question in column-I.



	Column-I		Column-II
(A)	Electric field due to $\frac{Q}{2}$ at C	(P)	Can be calculated
(B)	Electric field due to charge on outer surface of shell at O	(Q)	Can't be calculated
(C)	Modulus of electric potential due to charge on inner surface of shell at C	(R)	$\frac{kQ}{2R^2}$
(D)	Modulus of electric potential due to charge on outer surface of shell at A	(S)	$\frac{kQ}{4R}$
		(T)	$\frac{kQ}{2R}$

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

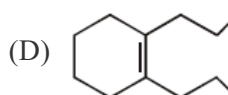
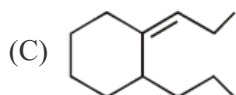
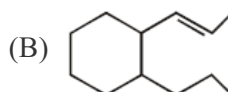
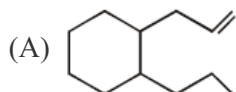
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

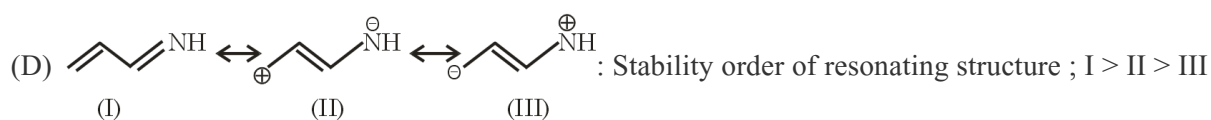
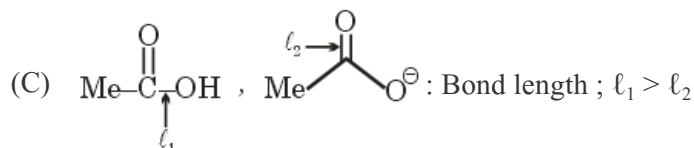
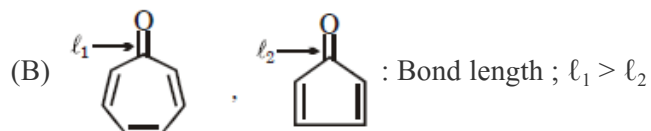
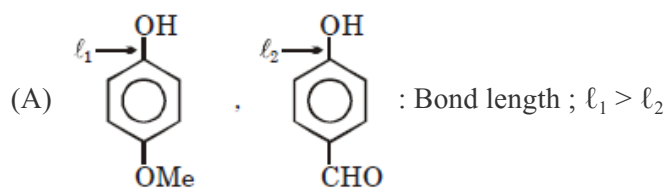
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Compounds which can show geometrical isomerism

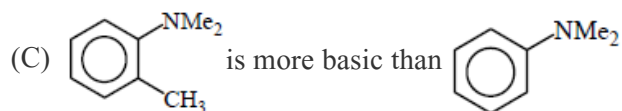
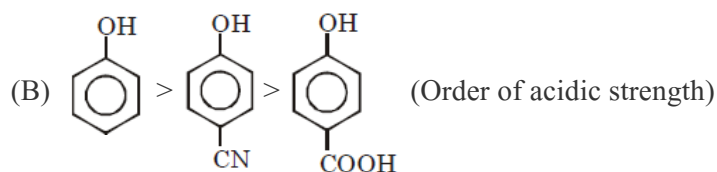


Space for Rough Work

2. Identify correctly matched options :



3. Select correct statement among following -



Space for Rough Work

4. In 20 ml 0.3 M BaCl_2 solution, 80 ml of 0.2 M H_2SO_4 solution is added. Select correct statement(s) :
 $(K_b)_{\text{H}_2\text{O}} = 1.8 \text{ K-Kg/mole}$
 [Take molality = molarity]
- (A) Boiling point of solution after addition of H_2SO_4 is 100.972°C
 (B) Boiling point of solution before addition of H_2SO_4 is 101.62°C
 (C) Boiling point of solution after addition of H_2SO_4 is 101.188°C
 (D) For an aqueous solution containing non-volatile, non-electrolyte solute, graph between ΔT_b and molality is a straight line passing through origin.
5. Two bulbs of volumes 200cm^3 and 100cm^3 , are connected by a short tube containing an insulating porous plug that permits equalization of pressure but not of temperature between the bulbs. The system is sealed at 77°C when it contains oxygen under a pressure of 1 bar. The small bulb is immersed at 27°C and the large bulb is placed at 127°C . Neglecting thermal expansion of the bulbs, select the correct options.
- (A) Final pressure inside the system would be greater than 1 bar
 (B) Final pressure inside the system would be less than 1 bar
 (C) Number of moles of O_2 would increase in small container compared to initial
 (D) Number of moles of O_2 would increase in big container compared to initial
6. Choose correct statement(s)-
- (A) In the balanced redox reaction : $x\text{As}_2\text{S}_3 + y\text{NO}_3^- + 4\text{H}_2\text{O} \rightarrow a\text{AsO}_4^{3-} + b\text{NO} + c\text{SO}_4^{2-} + 8\text{H}^+$
 molar ratio of $x : y = 3 : 28$.
 (B) When NH_4SCN oxidizes into SO_4^{2-} , CO_3^{2-} and NO_3^- , its equivalent weight will be $M/24$
 (C) When Bi_2S_3 converted into Bi^{5+} and S, its n-factor will be 7.
 (D) Equivalent weight of H_3PO_2 when it disproportionate into PH_3 and H_3PO_3 is $3M/4$

Space for Rough Work

7. Which of the following graphs is parabolic for fixed moles of an ideal gas at constant temperature in closed container
- (A) $\frac{1}{V^2}$ vs P (B) $\frac{P}{V}$ vs P (C) $\frac{P}{V}$ vs V (D) $\frac{P}{V}$ vs $\log V$
8. Borazine having/show.
- (A) Maximum number of atom in one plane is 12
 (B) Aromatic in nature
 (C) Has three different types of bond length data
 (D) All atoms lie in the nodal plane of π -electron cloud
9. Which of the following reaction(s) is/are **NOT** possible (stoichiometric coefficient of reactant is given) :-
- (A) $\text{BF}_3 + \text{F}^- \longrightarrow$ (B) $\text{PBr}_5 + \text{Br}^- \longrightarrow$
 (C) $\text{I}_2 + \text{I}^- \longrightarrow$ (D) $\text{B(OH)}_3 + 3\text{OH}^- \longrightarrow$
10. Which of the following statement is/are **INCORRECT**?
- (A) Graphite having dangling bond
 (B) In monoclinic sulphur S_6 units are predominately present
 (C) In molten state Iodine conduct electricity
 (D) Thermal conductivity of graphite is greater than diamond

Space for Rough Work

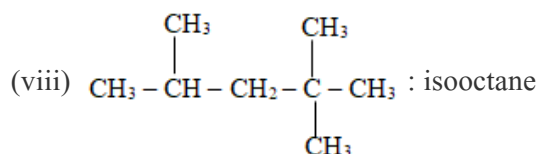
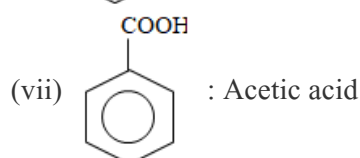
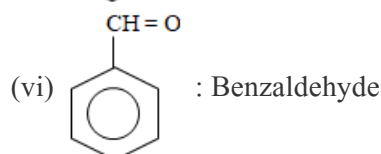
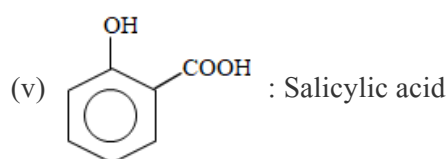
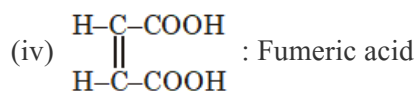
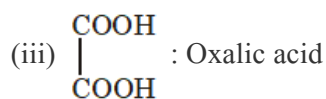
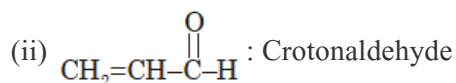
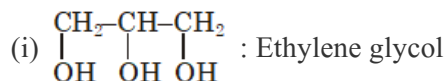
SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct answer is chosen.

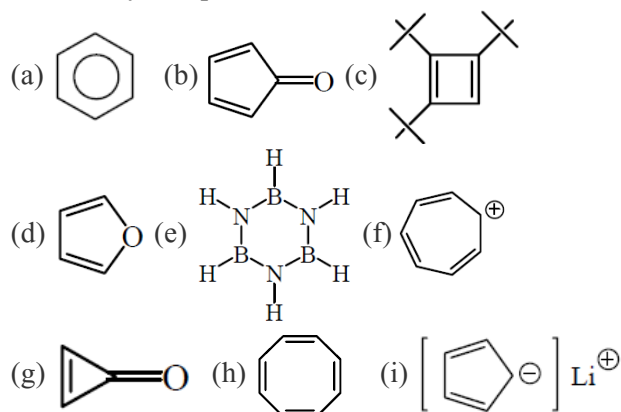
Zero Marks : 0 In all other cases.

1. Which of the following is/are incorrectly matched :

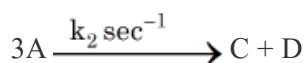
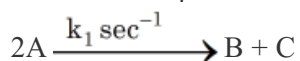


Space for Rough Work

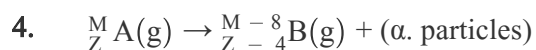
2. How many compounds are aromatic in nature



3. Consider two parallel reactions



if value of k_1 and k_2 are $x \text{ sec}^{-1}$ and $2x \text{ sec}^{-1}$ and 40 mole of A dissociate in time interval of t then mole of B produce in this time interval will be



(α . particles are helium nuclei, so will form He gas by trapping electron). The radioactive disintegration follows 1st order kinetics. Starting with 400 moles of A in a 16.4 litre closed flask, slope of P vs T graph after two half life is

($R = 0.082 \text{ litre atm / mol-K}$)

Space for Rough Work

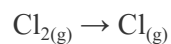
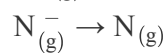
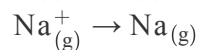
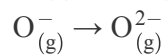
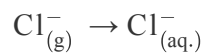
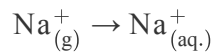
5. Predict the value of n in $AX_{(5-n)}L_{(n)}$ in linear geometry.

Where A = central atom

X = Halogen atom

L = Lone pair

6. Find the number of process which are exothermic.



Space for Rough Work

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen

Zero Marks : 0 In all other cases

1. Match the column

Column-I (Compound)		Column-II (Statements)	
(A)	Lactic acid	(P)	Soluble in aqueous NaHCO_3
(B)	Picric acid	(Q)	No COOH group is present
(C)	Squaric acid	(R)	Soluble in aqueous NaOH
(D)	Salicylic acid	(S)	$-\text{OH}$ group is present
		(T)	Liberates H_2 gas when treated with Na metal

2.

Column-I		Column-II	
(A)	$\text{NaClO}_4(\text{s})$	(P)	All the bond angle in anion is equal
(B)	$\text{PH}_4\text{I}(\text{s})$	(Q)	All the bond length in anion are equal
(C)	$\text{NaHCO}_3(\text{s})$	(R)	Having ionic interaction
(D)	$\text{Na}_2\text{HPO}_3(\text{s})$	(S)	Co-ordinate bond is present
		(T)	Hydrogen bond is present

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 40)

- This section contains **TEN (10)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If $al^2 + bm^2 + 2d\ell + 1 = 0$, where a, b, d are fixed real numbers such that $a - b = d^2$, then the line $\ell x + my + 1 = 0$ touches the fixed circle (where $b < 0$ and $\ell, m \in \mathbb{R}$)
 - (A) which cut the x-axis orthogonally
 - (B) with radius equal to $\sqrt{-b}$
 - (C) on which length of the tangent from origin is $\sqrt{d^2 + b}$
 - (D) with centre $\equiv (d, -b)$
2. If a, b, c are positive integers forming first three terms of increasing geometric progression with common ratio 'r', where $b - a$ is perfect cube and $\log_6 a + \log_6 b + \log_6 c = 6$. Let S denotes sum of 10 terms of another G.P. whose first term is 'a' and common ratio is $\frac{5r}{2}$, then
 - (A) $a + b + c = 189$
 - (B) $a + b + c = 273$
 - (C) sum of digits of S is 90.
 - (D) $S = 10^{10} - 1$

Space for Rough Work

3. In an isosceles $\triangle ABC$ with integral sides, $AB = 7$ cm and $\cos A = \frac{2}{7}$, then which of the following is/are true
- (A) Area of $\triangle ABC$ is $6\sqrt{5}$ cm² (B) Area of $\triangle ABC$ is $\frac{1029}{32}\sqrt{5}$ cm²
- (C) Circumradius of $\triangle ABC$ is $\frac{49\sqrt{5}}{30}$ cm (D) Circumradius of $\triangle ABC$ is $\frac{49\sqrt{5}}{12}$ cm
4. The curve $xy = c$ ($c > 0$) and the circle $x^2 + y^2 = 1$ touch at two points, then -
- (A) $c = 1/2$ (B) distance between points of contact is 2 unit
- (C) slope of tangent at point of contact is -1 (D) length of latus rectum of $xy = c$ is 2
5. If foci of an ellipse be $(-1, 2)$ and $(-2, 3)$ and its tangent at a point A is $2x + 3y + 9 = 0$
- (A) Length of the minor axis of the ellipse will be $2\sqrt{14}$
- (B) Co-ordinate of the point 'A' will be $\left(\frac{-32}{9}, \frac{-17}{27}\right)$
- (C) Distance between the foci is $2\sqrt{2}$
- (D) Product of the perpendiculars from foci to any tangent is 56
6. Consider number $N = 2^3 \times 3^2 \times 5^4$ then select the correct options(s)
- (A) Number of positive divisors of N is 60
- (B) Number of positive divisors which are divisible by 15 is 32
- (C) If $N = a \times b$, a, b are positive integers then ordered pair (a, b) is 60
- (D) Number of positive divisors of N is 24

Space for Rough Work

7. Let $N = 2^{200}({}^{200}C_0)({}^{200}C_{100}) - 2^{198}({}^{200}C_1)({}^{199}C_{99}) + 2^{196}({}^{200}C_2)({}^{198}C_{98}) - \dots + ({}^{200}C_{100})({}^{100}C_0)$ then
- (A) N is equal to the coefficient of x^{100} in $(1-x)^{200}$
- (B) The number of zeroes at the end of N is only one.
- (C) $N = 3^{101} \times k$ where k is an integer.
- (D) N is equal to the number of ways of distributing 200 different objects among 2 persons equally.
8. Let a variable circle touches a line, which is tangent to the parabola $y^2 = 4x$ at $(4, 4)$ and variable circle passes through focus of the parabola $y^2 = 4x$. If locus of its centre is conic 'C', then
- (A) directrix of conic 'C' is $x - 2y + 4 = 0$ (B) focus of conic 'C' is $(4, 0)$
- (C) latus rectum of conic 'C' is $2\sqrt{5}$ (D) axis of the conic 'C' is $2x + y - 2 = 0$.
9. Let from the point with abscissa 25, two tangents are drawn to the ellipse $24x^2 + 25y^2 = 600$ with foci at S_1 and S_2 . The points of contact of tangents are A and B. If the distance of A from S_1 is $\frac{60}{13}$ units, then
- (A) The distance of A from S_2 is $\frac{80}{13}$ units
- (B) The distance of B from S_1 is 5 units
- (C) The distance of B from S_2 is 5 units
- (D) The distance of A from the directrix corresponding to S_2 is $\frac{350}{13}$ units.
10. The equation $\sqrt{x+3-4\sqrt{x-1}} + \sqrt{(x+15)-8\sqrt{x-1}} = 2$ has
- (A) no real root (B) exactly one real root
- (C) exactly two real roots (D) infinite number of real roots

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If ONLY the correct answer is chosen.

Zero Marks : 0 In all other cases.

1. From a variable point $P(t^2, 2t)$ ($1 \leq t \leq 3$) on the parabola $y^2 = 4x$, perpendicular PM is drawn to the tangent at the vertex of the parabola. Now from the midpoint Q of PM, perpendicular QL is drawn to the focal chord of parabola through P. If the maximum length of QL is λ , then $[\lambda]$ equals (where $[.]$ denotes greatest integer function)
2. Consider the polynomials
 $P(x) = (x + \sqrt{2})(x^2 - 2x + 2)$
 $Q(x) = (x - \sqrt{2})(x^2 + 2x + 2)$
 $R(x) = (x^2 + 2)(x^8 + 16)$,
 then the coefficient of x^4 in $P(x) \cdot Q(x) \cdot R(x)$ is-
3. In a survey, 100 teenagers were interviewed to know their preferences for arrange marriage (AM), love marriage (LM), remain single (RS), and the collected data were as following
 $n(AM \cap LM \cap RS) = 10$, $n(AM \cap LM) = 20$
 $n(LM \cap RS) = 30$, $n(AM \cap RS) = 25$
 $n(\text{only AM}) = 25$, $n(\text{only LM}) = 5$, $n(\text{only RS}) = 8$ and some of them did not respond,
 then number of correct statements from the following is-
 (i) $n(AM) = 47$
 (ii) $n(LM) = 45$
 (iii) $n(RS) = 53$
 (iv) $n(AM \cup LM \cup RS) = 100$

Space for Rough Work

4. A ray of light travels along a line, strikes a parabola at $(-1, 1)$ and get reflected from it. Given that joint equation of incident and reflected ray is $x^2 - y^2 + 2x + 2y = 0$ and tangent at vertex of parabola is $x - y + 4 = 0$. If focus of parabola is (α, β) , then value of $|\alpha + \beta|$ is equal to
5. If $0 \leq x \leq 3\pi$, $0 \leq y \leq 3\pi$ and $\cos x \sin y = 1$, then possible number of ordered pair (x, y) is
6. The vertex A of $\triangle ABC$ is $(3, -2)$ and the equations to two of its medians are $5x + 3y = 11$ and $4x + 3y = 8$. If the equation of side BC is $ax + by = 7$, then $(a + b)$ is equal to
-

Space for Rough Work

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen

Zero Marks : 0 In all other cases

1.

Column-I		Column-II	
(A)	Circle passing through distinct points $(t^2, 1), (1, t), (2, 2t+1) \forall t \in \mathbb{R}$ passes through a fixed point (a, b) , then $a + b$ is	(P)	1
(B)	Tangent AB, AC are drawn from A to $x^2 + y^2 - 2x + 4y + 1 = 0$ meeting circle at B and C. If equation of circumcircle of $\triangle ABC$ is $x^2 + y^2 - x + y - 2 = 0$ such that A is (a, b) , then $a + b$ is	(Q)	3
(C)	The value of $\left(16 \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7} \right)$ is	(R)	5
(D)	Radius of a circle which is orthogonal to line pair $xy - 3x + 2y - 6 = 0$ and touches the line $x - 2y + 3 = 0$ is r then r^2 is	(S)	2
		(T)	7

Space for Rough Work

2.

Column-I		Column-II	
(A)	If the line $x - y - 2 = 0$ is rotated through an angle $\frac{\pi}{12}$ in the clockwise direction about point $P(5,3)$ and equation of line in new position is $x - \sqrt{3}y + k = 0$, then k is	(P)	Rational
(B)	The point $R\left(\frac{a}{b}, 0\right)$ lying on x-axis such that $(PR + QR)$ is minimum, if P and Q are $(1,1)$ and $(3,2)$ respectively, then value of $(a + b)$ is [where a and b are coprime natural number]	(Q)	Irrational
(C)	Let $A(3,4)$, $B(0,0)$ and $C(3,0)$ be vertices of ΔABC . If $P(\alpha, \beta)$ is a point inside the ΔABC , equidistant from sides of triangle, then $(\alpha + \beta)$ is	(R)	Even integer
(D)	Real value of x satisfying $x^4 - 2x^3 - x^2 - 2x + 1 = 0$ is	(S)	Odd integer
		(T)	Prime number

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work